

PYTHON - SE - BACKEND

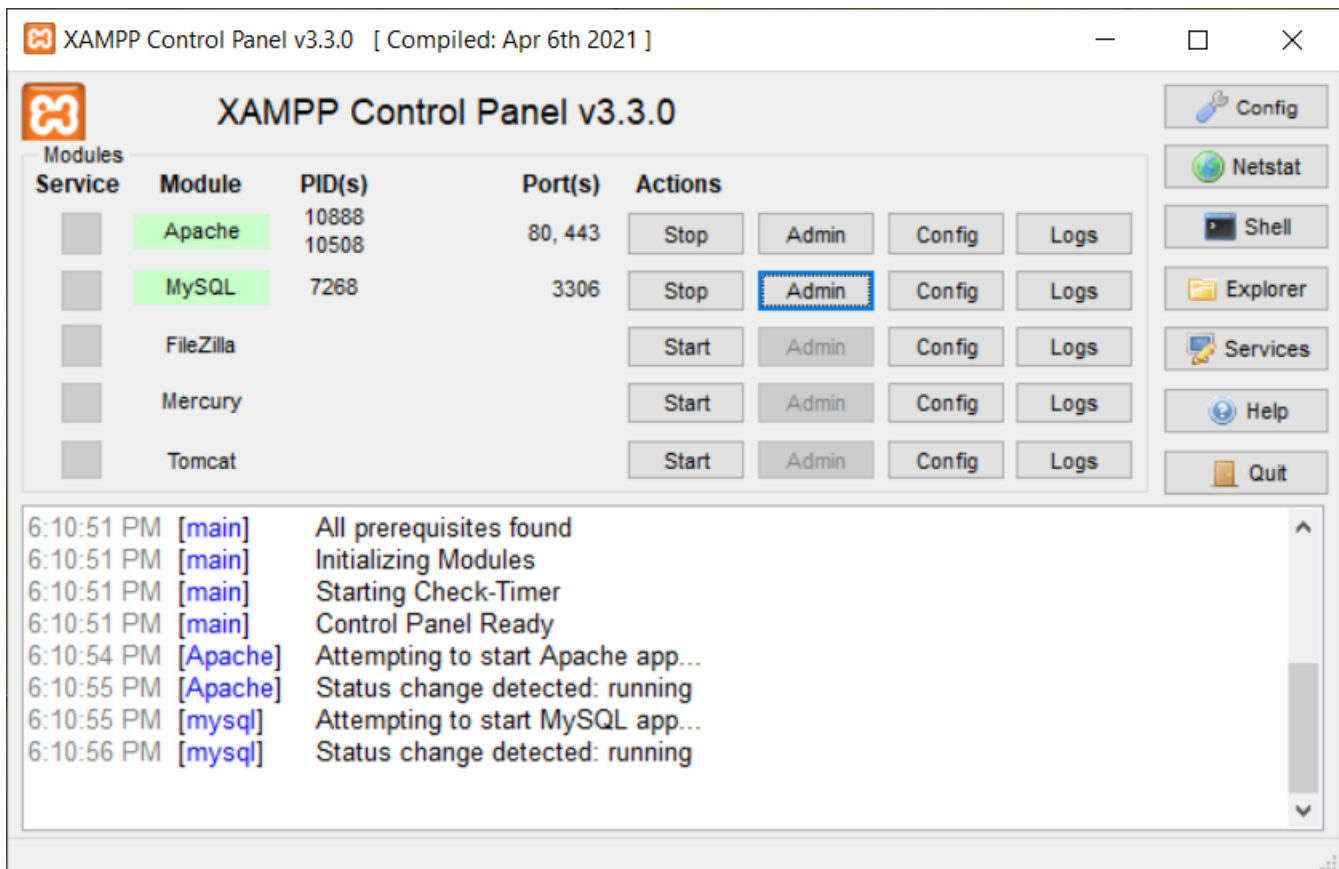
SOFTWARE ENGINEERING – MODULE 4

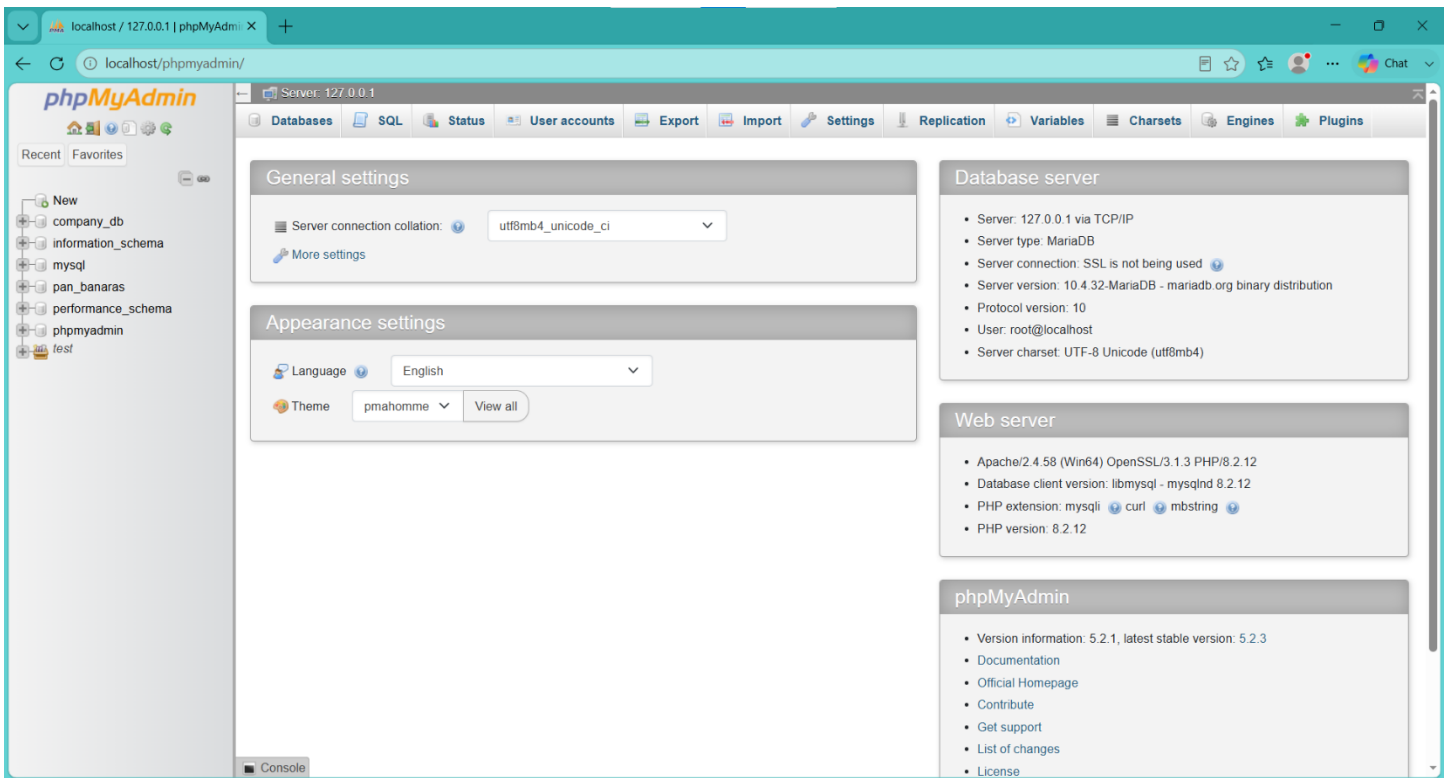
Database Management – SQL & PL/SQL

SESSION / ASSIGNMENT – 1

TASKS :

1. Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.





2. Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

➔ **CODE:**

```
CREATE TABLE Users(  
  user_id INT PRIMARY KEY,  
  username VARCHAR(100),  
  email VARCHAR(100),  
  followers_count INT );
```

OUTPUT:

 MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
SELECT * FROM `users`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

user_id	username	email	followers_count
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3. Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

→ CODE:

Use InstaClone;

```
INSERT INTO Users (user_id, username, email, followers_count)
```

```
VALUES
```

```
(1, 'dhruv_mistry', 'dhruv@gmail.com', 1250),
```

```
(2, 'travel_with_raj', 'raj@gmail.com', 3420),
```

```
(3, 'creative_isha', 'isha@gmail.com', 875);
```

```
SELECT * FROM Users;
```

OUTPUT:

The screenshot shows a MySQL database management interface with the following content:

- Server: MySQL-3306 » Database: instaclone » Table: Users
- Navigation tabs: Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations
- Query execution results:
 - MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)
 - USE InstaClone;
 - 3 rows inserted. (Query took 0.0005 seconds.)
 - Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)
- SQL queries executed:

```
USE InstaClone;
```

```
INSERT INTO Users (user_id, username, email, followers_count) VALUES (1, 'dhruv_mistry', 'dhruv@gmail.com', 1250), (2, 'travel_with_raj', 'raj@gmail.com', 3420), (3, 'creative_isha', 'isha@gmail.com', 875);
```

```
SELECT * FROM Users;
```
- Table data (Users):

user_id	username	email	followers_count
1	dhruv_mistry	dhruv@gmail.com	1250
2	travel_with_raj	raj@gmail.com	3420
3	creative_isha	isha@gmail.com	875
- Interface controls: Show all, Number of rows: 25, Filter rows: Search this table, Sort by key: None

4. Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

➔ **CODE:**

USE InstaClone;

```
CREATE TABLE Posts (  
    post_id INT PRIMARY KEY,  
    user_id INT,  
    caption VARCHAR(255),  
    post_date DATE,  
    FOREIGN KEY (user_id) REFERENCES Users(user_id)  
);
```

OUTPUT:

The screenshot shows a MySQL IDE interface with the following components:

- Toolbar:** Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines.
- SQL Editor:** Contains the SQL code:

```
1 USE InstaClone;  
2  
3 CREATE TABLE Posts (  
4     post_id INT PRIMARY KEY,  
5     user_id INT,  
6     caption VARCHAR(255),  
7     post_date DATE,  
8     FOREIGN KEY (user_id) REFERENCES Users(user_id)  
9 );
```
- Buttons:** Clear, Format, Get auto-saved query.
- Options:** ☐ Bind parameters.
- Footer:** Delimiter: ;, ☐ Show this query here again, ☐ Retain query box, ☐ Rollback when finished, ☒ Enable foreign key checks, Go.
- Hide query box** button.
- Execution Results:**
 - ✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)
 - USE InstaClone;
 - [Edit inline] [Edit] [Create PHP code]
 - ✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0073 seconds.)
 - CREATE TABLE Posts (post_id INT PRIMARY KEY, user_id INT, caption VARCHAR(255), post_date DATE, FOREIGN KEY (user_id) REFERENCES Users(user_id));
 - Console

5. Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

→ **1. MySQL**

MySQL is widely used for **web applications and websites**.

It is popular with PHP, WordPress, and other web development technologies.

It is suitable for small to large applications and is easy to use.

2. PostgreSQL

PostgreSQL is an advanced **open-source relational database**.

It is commonly used for web applications, data analysis, and applications requiring complex queries.

It is known for reliability, flexibility, and strong SQL support.

3. Oracle

Oracle Database is mainly used in **large enterprise systems**.

Banks, governments, and large organizations use it for handling huge amounts of critical data.

It provides strong security, scalability, and transaction management.

4. SQLite

SQLite is a lightweight database that does not require a separate server.

It is commonly used in **mobile apps, desktop applications, and embedded systems**.

It is especially useful when a simple, local database is required.